EX- 7: Missionaries Cannibal Problem

Program:

```
from collections import deque

def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)
    moves = [(1, 0), (2, 0), (0, 1),
             (0, 2), (1, 1)]

    queue = deque([(start, [start])])
    visited = {start}

    while queue:
        state, path = queue.popleft()
        m, c, boat = state

        if state == goal:
            for step in path:
                print(step)
            return

        for dm, dc in moves:
            if boat == 0:
                nm, nc, nb = m - dm, c - dc, 1
            else:
                nm, nc, nb = m + dm, c + dc, 0

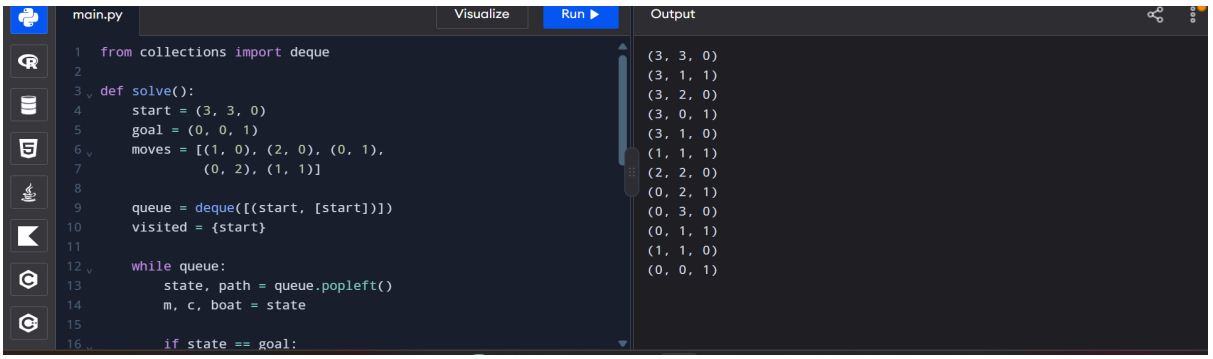
            if not (0 <= nm <= 3 and 0 <= nc <= 3):
                continue

            rm, rc = 3 - nm, 3 - nc

            if (nm == 0 or nm >= nc) and (rm == 0 or rm >= rc):
                new_state = (nm, nc, nb)

                if new_state not in visited:
                    visited.add(new_state)
                    queue.append((new_state, path + [new_state]))

solve()
```


main.py
Visualize
Run
Output
from collections import deque
def solve():
start = (3, 3, 0)
goal = (0, 0, 1)
moves = [(1, 0), (2, 0), (0, 1),
(0, 2), (1, 1)]
queue = deque([(start, [start])])
visited = {start}
while queue:
state, path = queue.popleft()
m, c, boat = state
if state == goal:
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)